

Computational Wind Engineering

Oscillators and aeroelasticity

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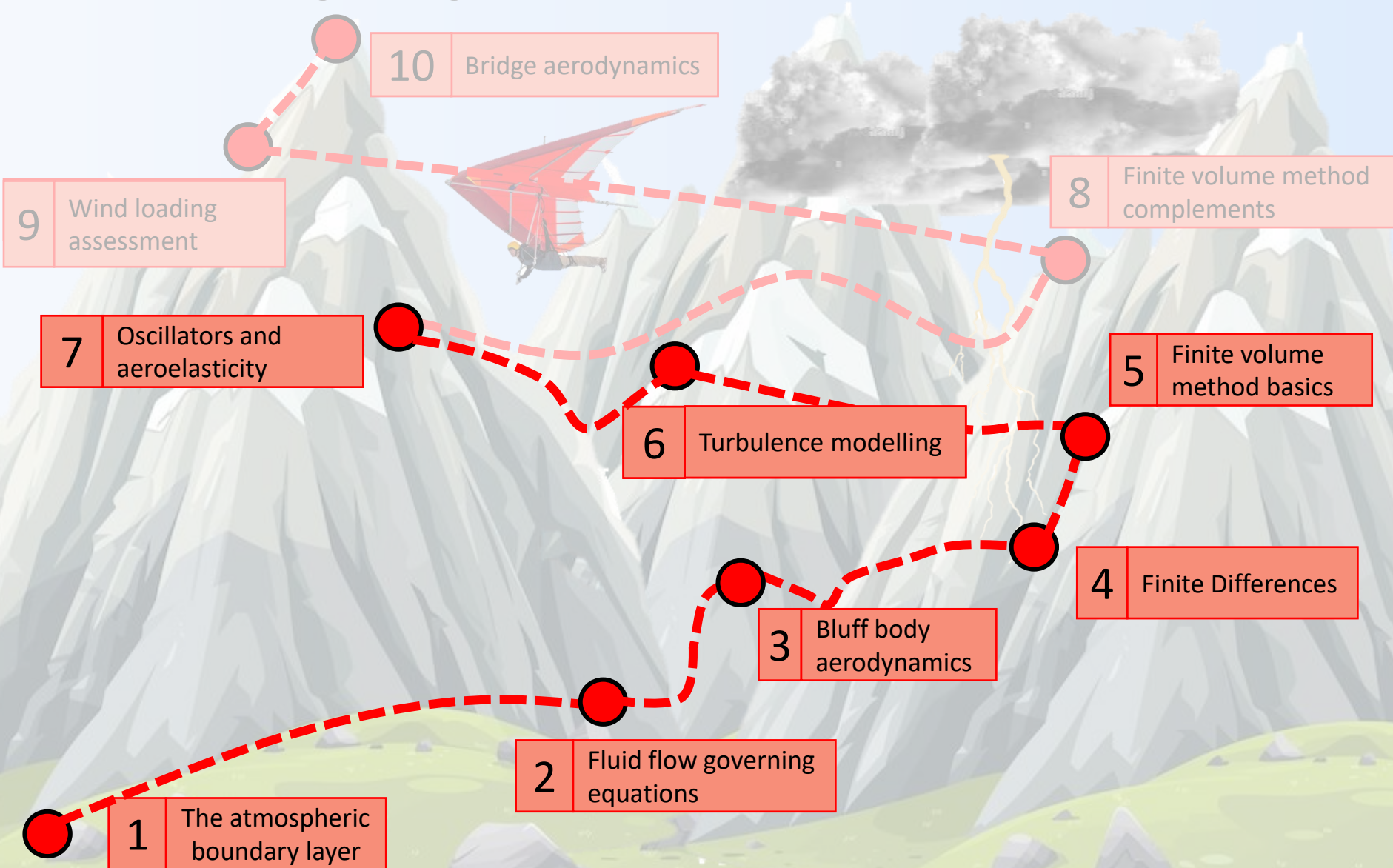
- Added mass and damping (along wind)
- Torsional divergence
- Galloping
- Vortex shedding
- Scruton number

Map of the course

Wind Engineering

Fluid Dynamics

Numerical methods



We now start considering the effect of wind on structures. **To this purpose, we firstly recall the simple oscillator.**

The governing equation is

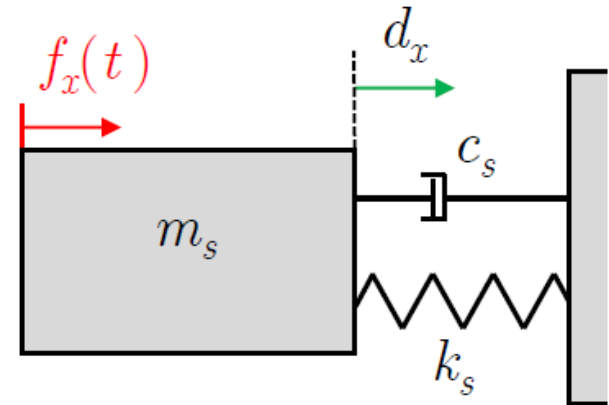
$$-m_s \frac{d^2 d_x}{dt^2} - c_s \frac{d d_x}{dt} - k_s d_x + f_x(t) = 0,$$

which can be rewritten as

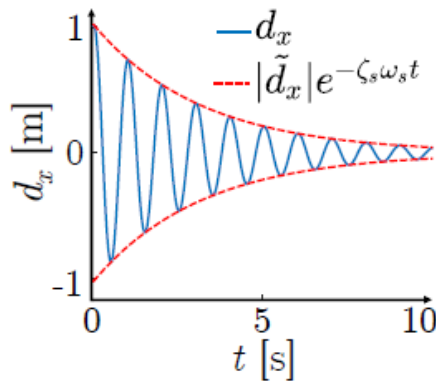
$$\ddot{d}_x + 2\zeta_s \omega_s \dot{d}_x + \omega_s^2 d_x = \frac{f_x(t)}{m_s},$$

with

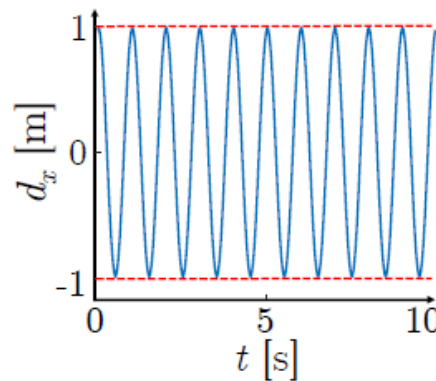
$$\omega_s = \sqrt{\frac{k_s}{m_s}}, \quad \zeta_s = \frac{c_s}{2\omega_s m_s},$$



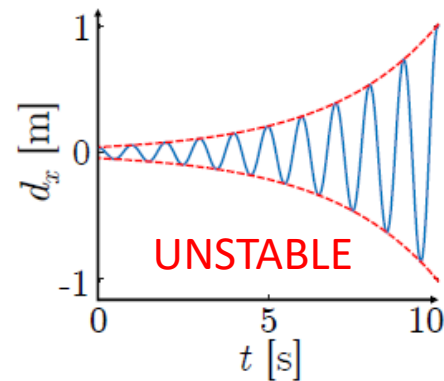
Assume the oscillator is at rest. If displaced and released, depending on the damping coefficient ζ_s **three different behaviours are possible**



$\zeta_s > 0$

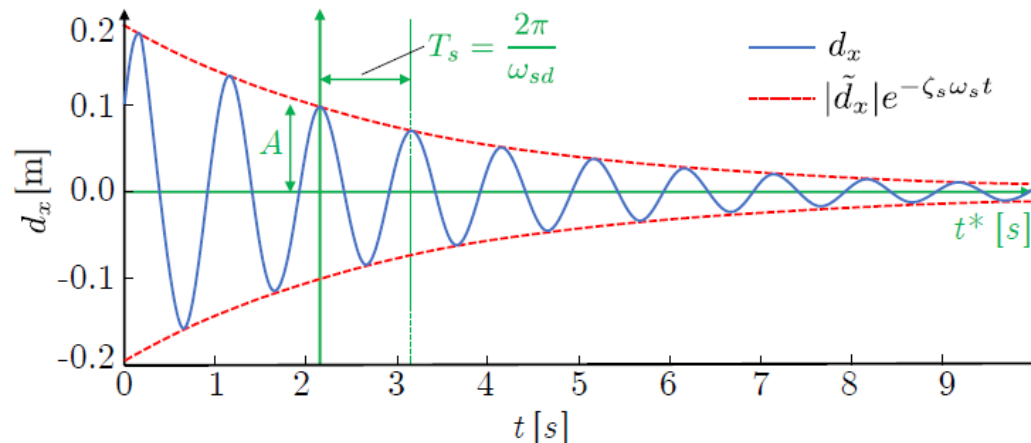


$\zeta_s = 0$

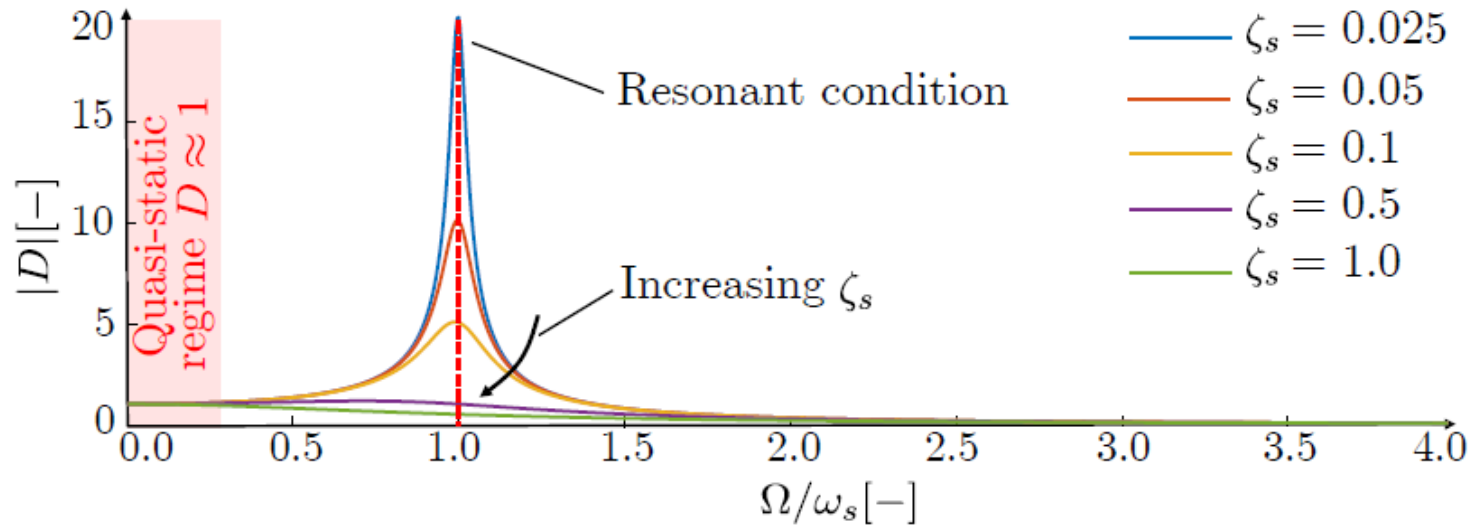


$\zeta_s < 0$

In particular, we have that



If the oscillator is forced periodically, the response looks as follows



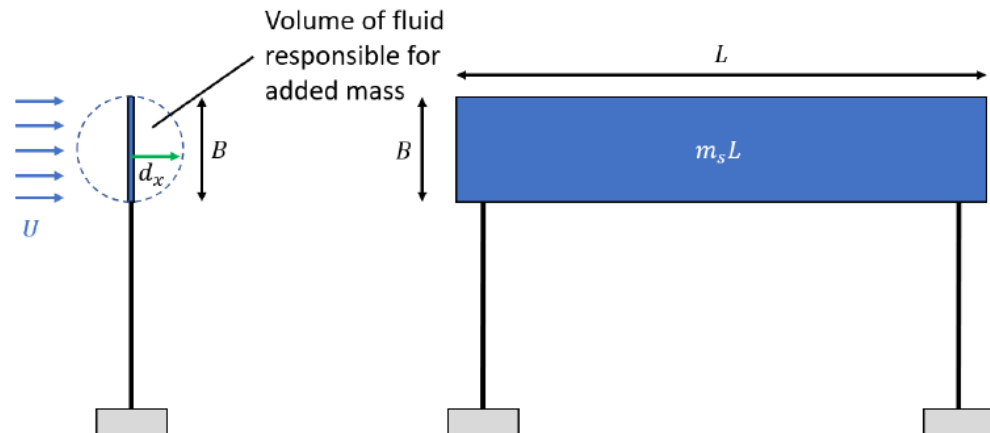
where D is the **dynamic amplification factor**, which evaluates how much the response is amplified with respect to the static one

$$|\tilde{d}_x| = |D(\Omega/\omega_s)| \frac{|\tilde{f}_x|}{k_s},$$

$$|D(1)| = \frac{1}{2\zeta_s},$$

Resonant condition

We now consider how oscillators (structures) might be affected by wind. Let us first consider the following case



The motion equation is

$$m_s \ddot{d}_x + c_s \dot{d}_x + k_s d_x = f_x(t).$$

Even without any wind, if the sign vibrates, **some air will move with it**. To consider the inertia of such air

$$m_a = \rho B^2 C_a,$$

where m_a is the **added mass**, $C_a = \pi/4$. The added mass is usually small in Wind Engineering applications

In order to calculate the aerodynamic force, we shall consider the **relative velocity between the sign and the fluid**. In particular

$$f_x(t) = \frac{1}{2}\rho C_d B (U - \dot{d}_x)^2,$$

so that the equation of motion now becomes

$$m_s \ddot{d}_x + c_s \dot{d}_x + k_s d_x = \frac{1}{2}\rho C_d B (U - \dot{d}_x)^2.$$

We can, however, regroup it as

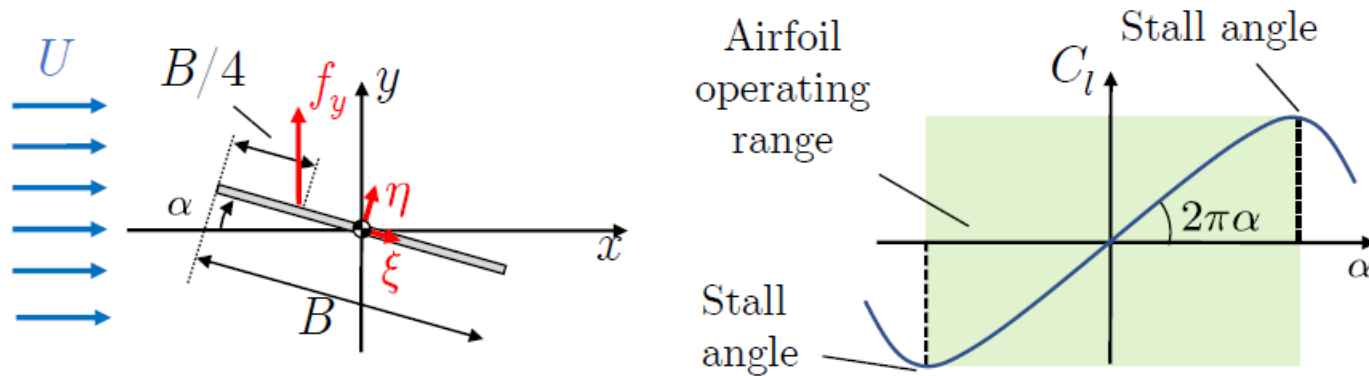
$$\frac{1}{2}\rho C_d B (U - \dot{d}_x)^2 = \frac{1}{2}\rho C_d B U^2 \left[1 - \frac{\dot{d}_x}{U} \right]^2 = \frac{1}{2}\rho C_d B U^2 \left[1 - 2\frac{\dot{d}_x}{U} + \left(\frac{\dot{d}_x}{U} \right)^2 \right]$$

and assuming $(\dot{d}_x/U) \ll 1$ we can write

$$m_s \ddot{d}_x + (c_s + \boxed{\rho C_d B U}) \dot{d}_x + k_s d_x = \frac{1}{2}\rho C_d B U^2.$$

$$c = c_s + \boxed{\rho C_d B U} = c_s + c_{ae}.$$

Let us now consider the following case (**thin airfoil**)



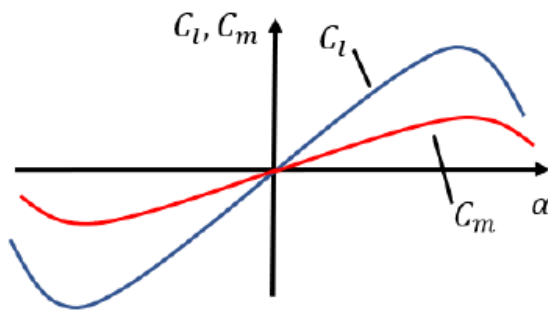
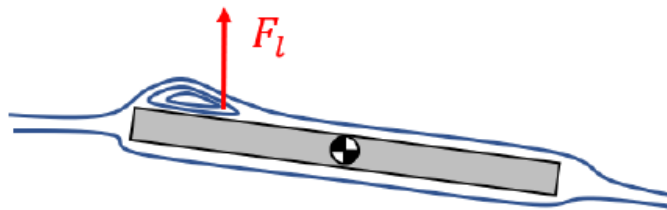
In the proximity of null angle of attack, we can express aerodynamic coefficients as

$$C_d \approx 0 \quad \text{and} \quad C_l(\alpha) = C_l(0) + 2\pi\alpha,$$

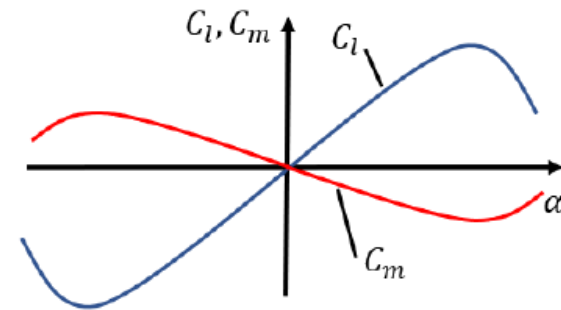
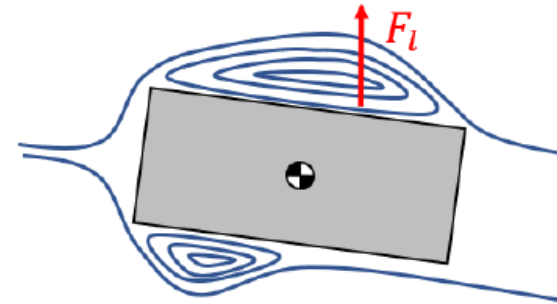
and considering that the lift is applied at $1/4B$

$$M(\alpha) = F_l \frac{B}{4} \cos(\alpha) = \frac{1}{2} \rho U^2 B C_l \frac{B}{4} \cos(\alpha) \approx \frac{1}{2} \rho U^2 B^2 \frac{\pi}{2} \alpha,$$

In a more general case, two scenarios are possible



(a)



(b)

In general, we can write

$$C_m = C_m(0) + \frac{dC_m}{d\alpha}\alpha.$$

The system governing equation thus becomes

$$m_s \ddot{\alpha} + c_s \dot{\alpha} + k_s \alpha = \frac{1}{2} \rho U^2 B^2 \left[C_m(0) + \frac{dC_m}{d\alpha} \alpha \right],$$

which can be rearranged as

$$m_s \ddot{\alpha} + c_s \dot{\alpha} + \left[k_s - \frac{1}{2} \rho U^2 B^2 C'_m \right] \alpha = \frac{1}{2} \rho U^2 B^2 C_m(0),$$

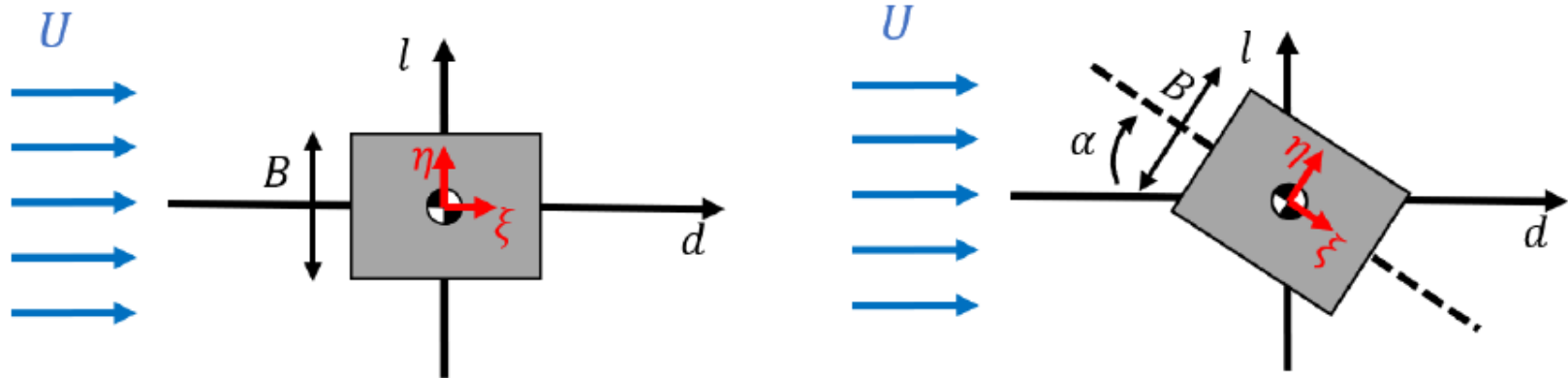
where $C'_m = \frac{dC_m}{d\alpha}$. The system stiffness is thus

$$k = k_s - \frac{1}{2} \rho B^2 U^2 C'_m = k_s + k_{ae},$$

being k_{ae} the **aeroelastic stiffness**. A critical condition is reached when **all the stiffness is lost, rendering equilibrium impossible**

$$k_s - \frac{1}{2} \rho U_{c,td}^2 B^2 C'_m = 0, \quad \Rightarrow \quad U_{c,td} = \sqrt{\frac{2k_s}{\rho B^2 C'_m}} = \sqrt{\frac{2m_s \omega_s^2}{\rho B^2 C'_m}}$$

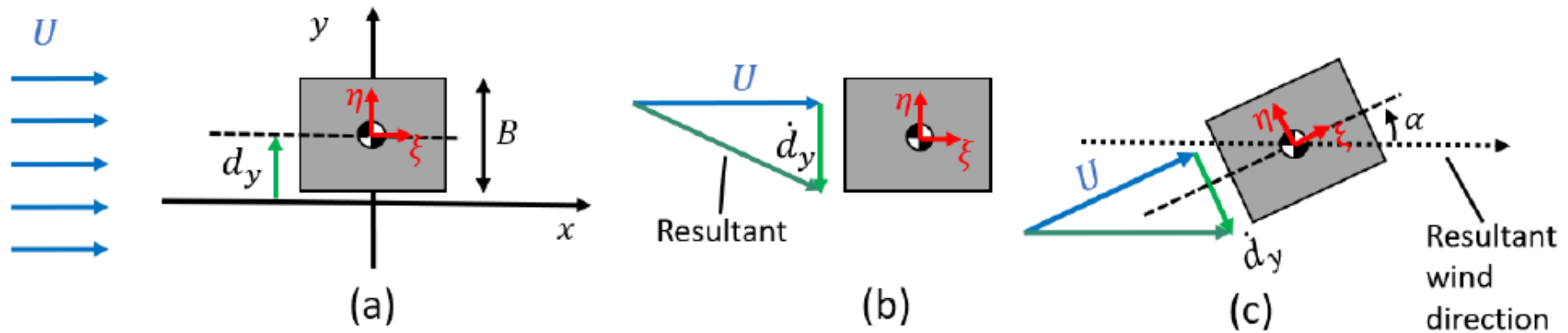
Before proceeding, it is necessary to clarify how aerodynamic coefficients measured in different reference systems relate. In wind tunnel tests, to modify the angle we have



$$\begin{bmatrix} F_d \\ F_l \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix} \begin{bmatrix} F_\xi \\ F_\eta \end{bmatrix} \quad \text{and similarly} \quad \begin{bmatrix} F_\xi \\ F_\eta \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{bmatrix} \begin{bmatrix} F_d \\ F_l \end{bmatrix}$$

so allowing to transform results in one coordinate system into the other one.

We now consider a system which allows for cross wind displacement, say d_y . Aerodynamic forces will be related to the **relative velocity** between the body and the fluid



$$\mathbf{u}_r = \mathbf{U} - \dot{\mathbf{d}},$$

and in particular assuming $\dot{d}_y \ll U$.

$$\alpha_e = \text{atan}(-\dot{d}_y/U) \approx -\frac{\dot{d}_y}{U}, \quad |\mathbf{u}_r| = \sqrt{U^2 + \dot{d}_y^2} \approx U.$$

Then, we can use a Taylor expansion

$$F_y = \frac{1}{2}\rho U^2 B [C_\eta(0) + C'_\eta \alpha_e]$$

so that

$$m_s \ddot{d}_y + c_s \dot{d}_y + k_s \alpha = \frac{1}{2}\rho U^2 B \left(C_\eta(0) - C'_\eta \frac{\dot{d}_y}{U} \right)$$

and reorganizing terms

$$m_s \ddot{d}_y + \left[c_s + \frac{1}{2}\rho U B C'_\eta \right] \dot{d}_y + k_s \alpha = \frac{1}{2}\rho U^2 B C_\eta(0).$$

Once more the system damping is the sum of the structural one and the aerodynamic one

$$c = c_s + \frac{1}{2}\rho U B C'_\eta = c_s + c_a,$$

A **critical condition** is reached when the total damping vanishes

$$\left[c_s + \frac{1}{2} \rho U B C'_\eta \right] = 0, \quad \longrightarrow \quad U_{c,g} = \frac{2c_s}{\rho B C'_\eta} = \frac{4\zeta_s \omega_s m_s}{\rho B C'_\eta},$$

We now only need to relate C'_η to C_d and C_l .

$$C_\eta = \sin(\alpha) C_d + \cos(\alpha) C_l$$

$$C'_\eta = \cos(\alpha) C_d + \sin(\alpha) C'_d - \sin(\alpha) C_l + \cos(\alpha) C'_l.$$

which evaluated for $\alpha = 0$ (the angle we used as base for the Taylor expansion) leads to

$$C'_\eta = C_d + C'_l,$$

Galloping is a perfect example to show how delicate aerodynamic phenomena are. In the case of the circular cylinder

$$C'_{\eta} = C_d.$$

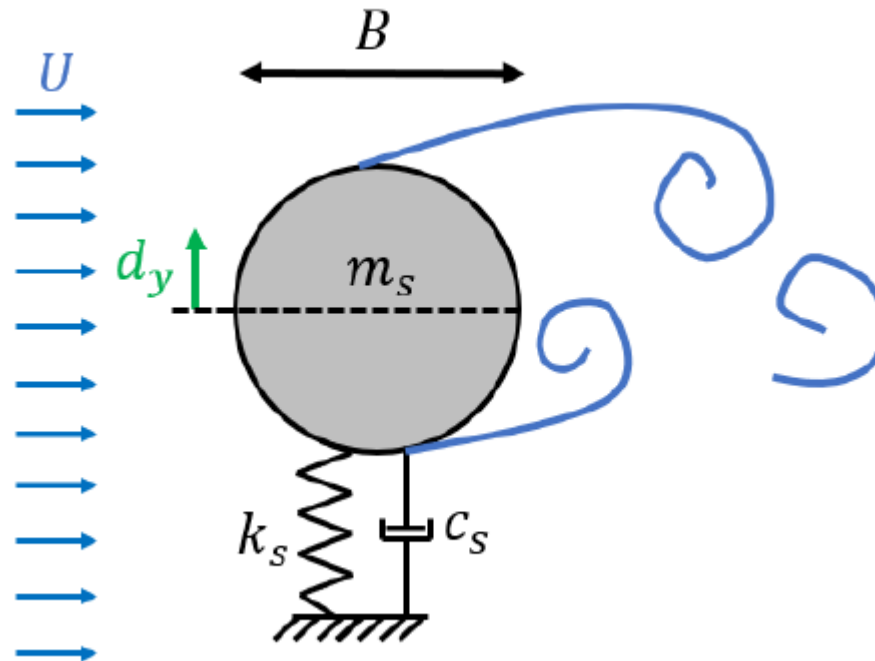
which is positive, and so excludes the possibility of galloping. **However, galloping or similar phenomena can develop in cables due to**

- Ice accretion
- Water rivulets due to rain.

In these cases, it is usually necessary to install dampers (e.g. stockbridge type).



We now wish to better understand vibrations induced by vortex shedding. We consider the following system



With an over-simplifying assumption, we might assume

$$f_{y,vs} = \frac{1}{2} \rho U^2 B C_{vs} \sin(\Omega_{vs} t),$$

The equation of motion becomes

$$\ddot{d}_y + 2\zeta_s \omega_s \dot{d}_y + \omega_s^2 d_y = \frac{1}{2m_s} \rho U^2 B C_{vs} \sin(\Omega_{vs} t).$$

where the critical condition in which Vortex Shedding **resonate** with the structure is reached at.

$$f_{vs} = St \frac{U_{c,vs}}{B} = f_s,$$

Remembering that at resonance $D(1) = 1/2\zeta_s$ we have

$$|\tilde{d}_{y,vs}| = \frac{|F_{y,vs}|}{k_s} D = \frac{1}{2} \rho U_{c,vs}^2 B C_{vs} \frac{1}{k_s} \frac{1}{2\zeta_s} = \frac{1}{2} \rho \left(\frac{f_s B}{St} \right)^2 B C_{vs} \frac{1}{m_s (2\pi f_s)^2} \frac{1}{2\zeta_s},$$

which can be reorganized as

$$\frac{|\tilde{d}_{y,vs}|}{B} = \frac{1}{16\pi^2} \frac{\rho B^2}{m_s \zeta_s} \frac{1}{St^2} C_{vs} \propto \frac{1}{Sc} \frac{1}{St^2} C_{vs}.$$

$$Sc = \frac{m_s \zeta_s}{\rho B^2}.$$

We, in fact, just introduced the **Scruton number**

$$Sc^* = \frac{m_s \zeta_s}{\rho B^2} \quad \text{or equivalently} \quad Sc = \frac{4\pi m_s \zeta_s}{\rho B^2}.$$

where the second definition above is often preferred for historical reasons. The model we analysed is actually too simple and, in reality, basing on wind tunnel tests, we can measure the function

$$\frac{|\tilde{d}_{y,vs}|}{B} = f(Sc),$$

In general, as a first check for cables

- if $Sc > 30$ the risk of significant VIV is modest;
- if $5 < Sc < 30$ the risk of VIV is significant and further studies shall be performed to evaluate their magnitude and impact;
- if $Sc < 5$ strong VIV are highly probable, potentially leading to strong serviceability and fatigue problems.